Apex Smart Structure v3.6

Build report: M5 / M15 version with timeframe presets, Student and Admin builds. Prepared by VConekt, 9 October 2026. Files: APEX_SMC_v3.6_STUDENT.pine (475 lines) and APEX_SMC_v3.6_ADMIN.pine (622 lines), Pine Script v6.

1. Summary

- **Why v3.6:** the teacher confirmed the strategy is traded on **M5 and M15**. The earlier defaults (v3.5) lost money on both timeframes in testing, so v3.6 adds tested settings for each one.
- **Timeframe preset:** one setting, **Auto** by default. It reads the chart timeframe and applies the M5, M15 or 30m / 1h settings.
- **How the settings were chosen:** 1,152 combinations tested on 2 years of gold. Chosen on **year 1 only**, then checked on **year 2**, which the choice never saw.
- **Result (gold, spread included):** M15 +0.22R per trade in both years. M5 +0.16R then +0.08R (weaker). About 3-4 signals a week each.
- **Visuals:** unchanged from v3.5. The Student build draws only BUY / SELL labels. The Admin build has every analysis visual as a toggle.
- **Status:** ready to paste. Not yet compiled in TradingView. Use **OANDA:XAUUSD** (needs volume data, see section 7).

Important: the results are from past data on one market. They show the presets beat the old settings; they do not guarantee future profit. The win rate is about 30% (fixed 3R target): most trades lose small, a few win big. Do not market win rates or returns from this report.

2. How the script follows the teacher's rules

The teacher provided two things: a **voice note** with the trading rules and a **Pine script** (sent together). Table A maps every rule in the voice note, table B maps every part of the teacher's script. The same logic runs in both builds; the Student build only hides the drawings.

#	A. Teacher's rule (voice note)	How v3.6 does it	Status
1	A Higher High is only valid once price pulls back: candles inside the peak candle's range do not count as a pullback.	The peak candle is tracked. Candles that stay inside its range are ignored; only **closes below the peak candle's low** count.	Followed
2	"Two or three candles" must close below the low of the peak candle to confirm the Higher Low.	Number of closes is set by the preset: **3 on M5, 2 on M15** (both inside the teacher's two-or-three range).	Followed
3	The Higher Low is the bottom of that pullback.	The lowest low of the confirmed pullback becomes the Higher Low (protected low).	Followed
4	Break of Structure (BOS) only when a candle breaks the previous high decisively.	BOS needs a candle **close** above the confirmed Higher High (a wick alone is not enough).	Followed
5	Trend changes when structure breaks the other way.	A close below the protected Higher Low flips the trend (CHoCH); everything then mirrors for sells (LH / LL, supply).	Followed
6	Look for zones inside the swing between the Higher Low and Higher High: Demand, Order Block, FVG.	Zones are collected while price moves; at each BOS only those **inside the new swing and not yet touched** become valid.	Followed

#	A. Teacher's rule (voice note)	How v3.6 does it	Status
7	If price makes a new high without retesting the old zones, those zones are no longer valid; use the new ones higher up.	On every new BOS all zones of the previous leg are **deleted** and only the new leg's zones are used.	Followed
8	When price pulls back after the BOS and retests a valid zone, give a BUY (SELL in a downtrend).	The first touch of a valid zone after the BOS gives BUY / SELL on the candle close. Each zone signals once; a close through it cancels it.	Followed
9	A buy can come at the 50% level of an FVG.	FVGs trigger at their 50% level.	Followed
10	Do not buy at every new zone; wait for the next valid level after the market retraces.	One trade at a time: zones touched while a trade is open are used up without a signal. The next valid zone is used on the next retrace.	Followed
11	Strategy is traded on M5 and M15 (confirmed at the end of the voice note).	**Auto preset** applies tested M5 and M15 settings from the chart timeframe.	Followed
12	Also use AVR (volume-based area), TBR and Mitigation Blocks (named in the voice note; not in the teacher's script).	Added as EVR (volume-climax candle), TBR (rejection candle) and Mitigation Block (last opposite candle before a break without an FVG). **Neither the voice note nor the teacher's script defines them**; these are our working definitions.	Our version
13	The teacher's own Pine script ("Apex Forex Academy - Smart Structure Indicator"), sent with the voice note.	Received and used as the starting point of every version (v3 to v3.6). Each part is mapped in table B below.	Followed

#	B. Teacher's Pine script: what it does	What v3.6 does	Status
1	Market structure from pivot points (3 candles each side).	Replaced by the voice-note rule (closes below the peak candle's low). Pivots appear 3 candles late and do not check that rule.	Replaced by voice-note rule
2	BOS when the close crosses the last pivot high / low.	BOS when a candle closes beyond the confirmed Higher High / Lower Low.	Kept, stricter
3	Demand: 1-3 red candles, then a strong green candle (supply = mirror).	Same pattern: 1-3 opposite candles, then a strong candle (body at least 1.5x the 20-candle average) closing beyond the base.	Kept
4	Order Block: bearish candle followed by bullish candles that leave an FVG (bearish = mirror).	Same idea: last opposite candle before the move that leaves the FVG; the move must close beyond it.	Kept
5	FVG: 3-candle gap.	Same 3-candle gap, minimum size 0.1 ATR, entry at the 50% level (voice note).	Kept + voice-note rule
6	Zone cancelled when price closes through it, or when structure flips.	Same, plus the voice-note rule: all old-leg zones deleted at every new BOS.	Kept + voice-note rule
7	Signal when price is inside a zone and closes beyond its edge, in the trend direction.	Signal on the first touch of a valid zone, on the candle close, in the trend direction. Each zone signals once.	Kept
8	RSI filter (BUY needs RSI 50+, SELL 50 or less), on by default.	Kept as an option, off by default: on a pullback RSI is usually below 50, so it blocked the retest buys the rules ask for.	Optional (off)
9	5-candle gap between signals; a zone could signal again and again.	One trade at a time and one signal per zone.	Replaced (cleaner)
10	Stop 50 pips beyond the zone, pip size fixed at 0.0001.	Stop 5 pips beyond the zone with automatic pip size (+0.5 ATR on M15). The fixed 0.0001 pip size was wrong for gold and JPY pairs.	Kept idea, fixed

#	B. Teacher's Pine script: what it does	What v3.6 does	Status
11	Target = last swing high / low, else 200 pips; target moved after a new BOS.	Fixed 3R on M5 / M15, swing high / low on 30m / 1h. The target no longer moves after entry.	Changed (tested)
12	Draws BOS labels, zone boxes, entry / SL / TP lines, TP / SL labels.	Student build shows only BUY / SELL (visual rules). Admin build keeps the analysis drawings as toggles.	Per visual rules
13	Alerts for BUY, SELL and BOS.	Same alerts; BUY / SELL alerts also carry entry, SL and TP.	Kept

Added by us, not in the voice note or the teacher's script (to protect students; each was tested and is part of the presets):

Addition	Why
Rejection candle required (BUY on a green close, SELL on a red close)	Stopped the script buying zone after zone in a falling market; fewer losing streaks in every test.
Stop loss at the zone edge (+0.5 ATR on M15) and a fixed 3R target on M5 / M15	The teacher gave no stop / target rule; these scored best in the year-1 / year-2 test.
New York session only on M5 (12:00-17:00 UTC)	M5 lost money in other hours in both years.
Pause after a stop-out on M15 (until the next BOS / CHoCH)	Avoids repeated buys while structure is failing; helped on M15.
Signals only on closed candles	Signals never appear and then disappear (no repaint).

3. The presets

Setting	M5 preset	M15 preset	30m / 1h preset
Picked automatically when the chart is	5 minutes or lower	6 to 15 minutes	above 15 minutes
Swing confirmation (candle closes)	3	2	2
Trading hours	**New York only, 12:00-17:00 UTC**	All day	All day
Take profit	Fixed 3R	Fixed 3R	Swing high / low (fallback 2R)
Stop loss	Zone edge + 5 pips	Zone edge + 5 pips **+ 0.5 ATR**	Zone edge + 5 pips
Pause after a stop-out	Off	**On**	Off
Rejection candle required	Yes	Yes	Yes
One trade at a time	Yes	Yes	Yes
Zone types	Demand / Supply, OB, FVG + EVR, TBR, Mitigation	Demand / Supply, OB, FVG + EVR, TBR, Mitigation	Demand / Supply, OB, FVG

The preset can also be set by hand (M5 / M15 / 30m-1h). The Admin build adds a **Manual** option that unlocks every setting individually.

4. The two builds

	STUDENT (give to students)	ADMIN (author only)
On the chart	Only BUY (green, below bar) and SELL (red, above bar) labels	BUY / SELL plus toggles: BOS / CHoCH, HH / HL / LH / LL, nearest-3 zone boxes, status panel (now shows the active preset), entry / SL / TP lines
Settings	One: Preset (Auto / M5 / M15 / 30m-1h)	Preset incl. Manual, plus every structure, zone, entry, session and risk setting
Hidden code	All unused drawing code removed (checked: nothing else can draw)	Not applicable
Alerts	BUY / SELL with entry, SL and TP in the message	Same, plus BOS / CHoCH alerts

5. How the settings were tested

- **Data:** gold (XAUUSD), 1-minute Dukascopy data, 1 Sep 2024 - 20 Sep 2026, built into 5m and 15m candles.
- **Costs:** 0.25 spread charged on every trade. Stop buffer measured the way TradingView does on gold (5 pips = 0.005).
- **Method:** the v3.6 rules were replayed in Python candle by candle. 576 combinations per timeframe of: swing closes (1/2/3), pause on/off, rejection on/off, session (all day / London+NY / NY), target (swing / 1.5R / 2R / 3R), extra stop buffer (0 / 0.5 ATR), zone set.
- **No peeking:** combinations were ranked on **year 1 only**. Year 2 was used only to check them. A setting was trusted only if it worked in both years **and** its neighbouring settings also worked (a broad plateau, not a lucky spike).

Timeframe	Combinations positive in year 1	Positive in year 2	What the winners had in common
M5	10%	13%	New York session only, rejection candle, fixed 3R, no extra stop buffer
M15	34%	78%	Fixed 3R (or 2R) target, stop +0.5 ATR, extra zone set; others could be either way

6. Results

Gold	Settings	Year 1	Year 2	Trades	Per week	Win rate
M5	Old v3.5 default	-0.18R	-0.15R	1,467	13.7	29%
M5	**v3.6 M5 preset**	**+0.16R**	**+0.08R**	383	3.6	30%
M15	Old v3.5 default	-0.16R	+0.16R	466	4.4	33%
M15	**v3.6 M15 preset**	**+0.22R**	**+0.23R**	350	3.3	31%
30m	30m / 1h preset (= v3.5)	+0.04R	+0.54R	237	2.2	35%
1h	30m / 1h preset (= v3.5)	-0.36R	+0.61R	117	1.1	34%

- "R" = result per trade in units of the risk taken. +0.22R means that, on average, each trade made 22% of what it risked, after spread.
- **M15 is the strongest result:** positive in both years and stable across neighbouring settings.
- **M5 is thin:** +0.08R in year 2 is close to break-even. Treat M5 as the weaker timeframe.
- 30m / 1h rows are split into first / second half of trades (earlier test), not calendar years. Both depend heavily on gold's strong rally in the second half (1h lost money in the first half).

7. Chart requirement: use a gold feed with volume

The EVR zone type needs volume. The tests used tick volume. On a chart without volume, EVR never fires and the results drop:

Preset	With volume (EVR on)	Without volume (EVR inactive)
M5	+0.12R overall	+0.03R (about break-even)
M15	+0.22R overall	+0.12R

Use OANDA:XAUUSD on TradingView, which provides tick volume. TVC:GOLD may not have volume.

8. Visual rules check

The v3.6 builds were run through the same automated drawing check as v3.5 (spec_audit.py):

- **Student:** only two drawing calls exist, the BUY and SELL labels. All other drawing code was removed and the removal was verified as pure deletion (no logic line changed).
- **Admin:** every visual is behind its own toggle; trade lines are deleted on TP / SL and before a new trade; only the 3 nearest zones are drawn.
- Everything is calculated on closed candles only, so signals do not repaint.

9. Open points

- **TradingView compile:** not yet done (needs a signed-in account). Paste the Student file, Add to chart on OANDA:XAUUSD 15m and 5m.
- **EVR / TBR / Mitigation definitions** are ours: the voice note names them but neither it nor the teacher's script defines them. They help on M5 / M15 in testing; if the teacher defines them differently, the presets must be re-tested.
- **One market only:** presets were tested on gold. Other symbols are untested.
- **Win rate ~30%:** students must expect more losing trades than winning ones; the edge comes from 3R winners.
- **Voice note:** the M5 / M15 confirmation comes from the end of the teacher's voice note (confirmed by Khakan). The full audio was not re-transcribed.

10. Next steps

#	Action	Owner
1	Paste APEX_SMC_v3.6_STUDENT.pine into Pine Editor, Add to chart on OANDA:XAUUSD M15 and M5, report any error	VConekt
2	Check signals against the teacher's own chart reading on a few recent days	VConekt + Academy
3	Confirm what AVR, TBR and Mitigation Block mean (the only rules not in the voice note or script)	Academy
4	Re-test the presets if those definitions differ from ours	VConekt
5	Forward-test on a demo account for 4-6 weeks before showing any results to students	Academy

Files: APEX_SMC_v3.6_STUDENT.pine, APEX_SMC_v3.6_ADMIN.pine; built by build_v36.py + strip_student.py; tests in m5m15_grid.py and v3_replay.py; this report by make_v36_report.py. Educational charting tool. Not financial advice.